

Education

Military Institute of Science and Technology (MIST) — *M.Sc. Engg in Computer Science and Engineering* Apr 2026 – Present

Southeast University — *BSc in Computer Science and Engineering* Oct 2021 – Oct 2025
CGPA: 3.87/4.0

Relevant Coursework: Artificial Intelligence, Image Processing, Data Mining, Research Methodology

Experience

Research Assistant — *Islamic University of Technology (IUT), OIC* Gazipur, BD
 Nov 2025 – Present

- Architected decentralized Deep Learning and Federated Learning frameworks to address privacy constraints in large-scale data analysis.
- Developed novel algorithms for Causal Inference to improve auditing transparency and predictive accuracy in high-dimensional datasets.
- Five Q1 journal acceptances proposing new methodologies in decentralized AI and causal inference.

Research Assistant — *CAIR, Southeast University* Dhaka, BD
 Jan 2025 – Oct 2025

- Designed advanced Computer Vision architectures (ViT, Hybrid TransUNet, Cross-Modal networks) for medical image analysis, integrating XAI methods (SHAP/LIME) and attention mechanisms for interpretable clinical decision support.
- Performed high-precision segmentation and classification on complex MRI and X-ray datasets to enhance diagnostic accuracy.
- Two papers accepted at IEEE COMPAS 2025; two Q1 journal manuscripts and one conference paper under review.

Selected Publications

MangoLeafNet-XAI: An Attention-Enhanced Deep Learning Architecture for Accurate and Interpretable Mango Leaf Disease Classification [Paper]

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Farzan Majeed Noori, Md Zia Uddin, Abdul Kadar Muhammad Masum
 Frontiers in Plant Science [Q1] [🔗](#)

Integrating Domain-Specific Convolutional Features with Swin Transformers for Transparent Cervical Cancer Diagnostics [Paper]

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Md Fokrul Islam KHAN, Md Zia Uddin, Ulysse Teller Masao Côté-Allard, Abdul Kadar Muhammad Masum
 Frontiers in Oncology [Q2] [🔗](#)

FedMal-XAI: An Explainable Federated Vision Transformer Leveraging Knowledge Distillation for Privacy-Preserving Malaria Detection [Paper]

Md. Tofael Ahmed Bhuiyan, **Md. Abdur Rahman**, Md Fokrul Islam Khan, Farzan M. Noori, Abdul Kadar Muhammad Masum
 Frontiers in Public Health [Q1] [🔗](#)

DFU-GCNet: A Global Context-Enhanced Inception Network for Robust and Interpretable Diabetic Foot Ulcer Classification [Paper]

Md. Tofael Ahmed Bhuiyan, **Md. Abdur Rahman**, Farzan M. Noori, Abdul Kadar Muhammad Masum
 Frontiers in Digital Health [Q1] [🔗](#)

Privacy-Preserving ESG Risk Prediction via Federated Hybrid Neural-Boosting Ensembles [Paper]

Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, **Md. Abdur Rahman**, Md. Tofael Ahmed Bhuiyan, Ramona Birău, Virgil Popescu
 IEEE Access [Q1] [🔗](#)

Predicting Corporate Bankruptcy Through a Scalable and Interpretable Knowledge Distillation Framework [Paper]

Md. Abul Kalam Azad, Abdul Kadar Muhammad Masum, **Md. Abdur Rahman**, Md. Tofael Ahmed Bhuiyan, Thanh Lam Nguyen, Hasan Dincer, Serhat Yuksel
 Discover Artificial Intelligence [Q1] [🔗](#)

- EcoStack-Pro: An Adaptive Federated Learning Framework for Interpretable ESG Auditing Across Heterogeneous Industrial Sectors** [Paper]
Md. Abul Kalam Azad, Abdul Kadar Muhammad Masum, *Md. Abdur Rahman*, Md. Tofael Ahmed Bhuiyan, Farzan M. Noori, Md Zia Uddin
[Frontiers in Artificial Intelligence \[Q1\]](#) [↗](#)
- Bridging Data Silos in Corporate Governance: A Hierarchical Stacking Ensemble with Federated Dynamic Aggregation** [Paper]
Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*
[Applied Computational Intelligence and Soft Computing \[Q1\]](#) [↗](#)
- Beyond Short-Term Noise: Robust Long-Horizon Bankruptcy Prediction Using Gated Dual-Branch Transformers** [Paper]
Rezwanul Huque Khan, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Faisal Md. Abdur Rahman, Abdul Kadar Muhammad Masum
[Applied Computational Intelligence and Soft Computing \[Q1\]](#) [↗](#)
- Do LLMs Distinguish Between Halal and Haram? Benchmarking Islamic Cultural Alignment in General vs. Arabic-Centric SLMs** [Paper]
Md Abdur Rahman, Md Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
[6th Muslims in ML Workshop, ICML 2026](#) [↗](#)
- Tokenizer Transplantation: Mitigating Autoregressive Collapse in Edge-Efficient Bengali ASR** [Paper]
Sanjid Hasan, *Md Abdur Rahman*
[6th Muslims in ML Workshop, ICML 2026](#) [↗](#)
- SFL-MTKD++: Split ViT Federated Learning with Adaptive Multi-Teacher Distillation for Interpretable Malaria Diagnosis** [Paper]
Md Tofael Ahmed Bhuiyan, *Md Abdur Rahman*, Abdul Kadar Muhammad Masum
[6th Muslims in ML Workshop, ICML 2026](#) [↗](#)
- AlzFed-XAI: High-Fidelity Interpretable Alzheimer’s Diagnosis with Privacy-Preserving Federated Learning** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
[5th Muslims in ML Workshop, NeurIPS 2025](#) [↗](#)
- FAKD-XAI: Feature-Aligned Knowledge Distillation with Explainable AI for Efficient Brain Tumor Classification** [Paper]
Md. Abdur Rahman, Sabik Aftahee, Md Ashiqur Rahman, Lamim Zakir Pronay
[ICDSAIA 2025](#) [↗](#)

Under Review Publications

HA-DynAU-Net: A Hybrid Attention Dynamic Aware U-Net for Precise Skin Lesion Segmentation
(1st Author) — Submitted to IEEE Open Journal of the Computer Society

Honors & Professional Activities

- **3rd Position (Poster Presentation)**, Student Research Day 2025 [Certificate]
- **Champion**, Fragments of Feeling: A Contextual Emotion Challenge (2025) [GitHub]
- **1st Runners-up**, CalorieQuest: Predicting Energy Burn with Data (2025) [GitHub]
- **2nd Runners-up**, DU AI Challenge (2025) [GitHub]
- **2nd Runners-up**, REACT 2026 Datathon, hosted by IEEE SEU Student Branch [GitHub]
- **Finalist (Top 10)**, CUET ETE Datathon 2025 [GitHub]
- **Program Committee**, 21st BEA Workshop co-located with **ACL 2026**

Professional Certification

- **AWS Certified Cloud Practitioner**, Amazon Web Services Dec 2025 – Dec 2028
- **GitHub Foundations Certification**, GitHub Nov 2024 – Nov 2027

Skills

Languages: C, C++, Java, Python

Frameworks: Pandas, NumPy, Matplotlib, Seaborn, PyTorch, HuggingFace, OpenCV, Django, FastAPI, Spring Boot

Technologies: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Git/GitHub, Cloud Computing, LaTeX